

Seat No.: _____

Enrolment No. _____

NATIONAL FORENSIC SCIENCES UNIVERSITY
M.Sc. Digital Forensics & Information Security
Semester –II , September 2024

Subject Code: CTMSDFIS SII P4
Subject Name: Artificial Intelligence
Time: 2pm to 5pm

Date: 27/9/2024

Total Marks: 100

Instructions:

1. Write down each question on separate page.
2. Attempt all questions.
3. Make suitable assumptions wherever necessary.
4. Figures to the right indicate full marks.

Q.1

Attempt any three.

Marks

- (a) How do you use lists and dictionaries in Python? What's the difference between a list and an array in Python? **08**
- (b) In a group of 60 students, 6 were randomly chosen, and their final assessment marks were recorded as follows: **80, 85, 90, 75, 95, and 70**. Find the **mean, median, variance**, and **standard deviation** of their marks. **08**
- (c) Explain types of Learning with proper Example. **08**
- 1) Supervised 2) Unsupervised
3) Reinforcement 4) Semi-supervised
- (d) Do products of given Matrix and Vector **08**

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$$

Q.2

Attempt any three.

- (a) Given the following **confusion matrix** for a binary classification model: **08**

	Predicted Positive	Predicted Negative
Actual Positive	40	10
Actual Negative	5	45

- (i) Calculate the **Accuracy, Precision, Recall**, and **F1-Score** for this model.
- (ii) Briefly explain the significance of each of these metrics in evaluating a classification model.
- (b) What is the Decision Tree algorithm and how does it work in supervised learning? **08**

- (c) Define underfitting and overfitting in machine learning models. Discuss the techniques to prevent these issues and improve model performance. 08
- (d) Explain working of K-means & Hierarchical clustering. Discuss about pros and cons of both unsupervised algorithms. 08

- Q.3** **Attempt any three.**
- (a) Describe the architecture of an Artificial Neural Network (ANN). Discuss how neurons, activation functions, and layers contribute to its data processing. Illustrate the architecture with a diagram. 08
- (b) What is a perceptron neural network? Calculate the output of a perceptron with five inputs and a bias term using the input values (0.3, 0.6, 0.8, 0.4, 0.7) and weights [0.1, -0.2, 0.4, 0.3, -0.1, 0.5]. Apply the ReLU activation function to determine the final output. 08
- (c) Explain the structure of a Convolutional Neural Network (CNN). Highlight the role of convolutional layers, pooling layers, and fully connected layers in image processing tasks. 08
- (d) What is a Recurrent Neural Network (RNN)? Describe its architecture and how it processes sequential data. What are some common applications of RNNs, and what challenges can occur during training? How can these challenges be addressed? 08

- Q.4** **Attempt any two.**
- (a) Explain difference between object detection, object localization, object recognition, object segmentation. Write steps to detect face from given image using machine learning classifier. 07
- (b) Explain the role of pattern recognition in computer vision. How is it applied to real-world scenarios such as face detection or handwriting recognition? 07
- (c) What is clustering in NLP, and how is it used to group similar text? What are some different methods of clustering text data? 07

- Q.5** **Attempt any two.**
- (a) What are the different machine learning algorithms used for detecting malware? How do these algorithms work to classify and identify malware? 07
- (b) How does machine learning improve anomaly detection systems in cybersecurity? What are the advantages and drawbacks of machine learning-based anomaly detection compared to traditional rule-based methods? 07
- (c) How do intrusion detection systems (IDS) that use machine learning operates? How do machine learning-based IDS differ from traditional IDS approaches? 07

Seat No.: _____

Enrolment No. _____

NATIONAL FORENSIC SCIENCES UNIVERSITY
M.Sc. Digital Forensics and Information Security
Semester – II – July - 2024

Subject Code: CTMSDFIS SII P2**Date:** 11/07/2024**Subject Name:** Web Application Security**Time:** 02:30 PM to 05:30 PM**Total Marks:** 100**Instructions:**

1. **Write down each question on a separate page.**
2. **Attempt all questions.**
3. **Make suitable assumptions wherever necessary.**
4. **Figures to the right indicate full marks.**

		Marks
Q.1	Answer the following question (Attempt any three)	24
	(a) Discuss CVE and how does pen-tester will use CVE concepts during web security pen-testing.	08
	(b) Discuss the concept of cookies and session and also write four steps to stop the attacks to steal the cookies and hijacking the session of web application.	08
	(c) Discuss Cross Site Scripting with example.	08
	(d) Discuss docker container security.	08
Q.2	Answer the following question (Attempt any three)	24
	(a) Discuss CVSS and how does pen-tester will use CVSS during web security pen-testing.	08
	(b) Discuss the concept of proxy, interception, intruder and HTTP history and also elaborate its use during web application security pen-testing.	08
	(c) Discuss SSRF with example and how does it different from CSRF.	08
	(d) Discuss the concept of un-known vulnerability and false positive. As web application security pen-tester, how do you tackle with false positive during the pent-testing process.	08
Q.3	Answer the following question (Attempt any three)	24
	(a) Discuss STRIDE with example.	08
	(b) Discuss the concept of repeater and decoding and how does it is helpful to pen-tester during web security pent-testing.	08
	(c) Discuss Security Mis-configuration and CSRF with example.	08
	(d) Discuss the concept of API and API security.	08
Q.4	Answer the following question (Attempt any Two)	14
	(a) Discuss the Vulnerability Assessment Life cycle.	07
	(b) Discuss the concept of injection and discuss two injection attacks with example.	07

	(c)	Discuss the CMS security	07
Q.5		Answer the following question (Attempt any Two)	14
	(a)	Discuss the concept of origin and encoding and how does developer use it as defense mechanism to fight against web application attacks.	07
	(b)	Write note on authentication and authorization bypass with example.	07
	(c)	Discuss subdomain and virtual host enumeration techniques with example.	07

--- End of Paper---

NATIONAL FORENSIC SCIENCES UNIVERSITY

Semester End Examination (December – 2024)
M.Sc. Digital Forensic and Information Security
Semester - III

Subject Code: CTMSDFIS SHI P1
Subject Name: Network Security and Forensics
Time: 10.30am to 1.30pm

Date: 04.12.2024**Total Marks: 100****Instructions:**

1. Write down each question on a separate page.
2. Attempt all questions.
3. Make suitable assumptions wherever necessary.
4. Figures to the right indicate full marks.

			Marks
Q.1		Attempt any three.	
	(a)	Discuss the steps involved in a wireless network penetration test. How do sniffing tools assist in this process?	08
	(b)	Explain Substitution and transposition techniques in cryptography. Provide examples for each.	08
	(c)	Discuss the importance of Port scanning and its role in vulnerability assessment.	08
	(d)	Discuss the tools used for network traffic analysis and their significance in forensic investigation.	08
Q.2		Attempt any three.	
	(a)	Compare and discuss ICMP and TCP scanning techniques. Explain their use.	08
	(b)	Discuss the role of promiscuous mode and monitoring modes in wireless network penetration testing.	08
	(c)	Illustrate, how phishing attacks are conducted and suggest countermeasures to prevent them.	08
	(d)	Discuss the methodology of investigating network based digital evidence.	08
Q.3		Attempt any three.	
	(a)	You are given network logs showing unusual traffic and failed login. How would you use network forensic tools to identify the possible attack?	08
	(b)	Explain the 802.11 protocol family and the security enhancements introduced in newer standards.	08
	(c)	Discuss the principles of symmetric and asymmetric encryption. Explain it with example.	08
	(d)	Evaluate the role of SIEM in identifying and mitigating network threats.	08
Q.4		Attempt any two.	
	(a)	Discuss the working of Public Key Infrastructure (PKI) and its significance.	07
	(b)	Describe 7 layers of OSI Model, highlighting specific role of each layer in data transmission.	07

	(c)	Describe the process of ARP poisoning. What steps are involved in executing such attack.	07
Q.5		Attempt any two.	
	(a)	Compare and contrast WEP, WPA, WPA2, and WPA3 in terms of security features.	07
	(b)	Explain how hash functions are used to ensure data integrity.	07
	(c)	Explain all hardware devices are used in networking.	07

--- End of Paper---

Seat No.: _____

Enrolment No. _____

NATIONAL FORENSIC SCIENCES UNIVERSITY

M. Tech. Cyber Security - Semester - I - DEC-2024

Subject Code: CTMTCS23 SI P1

Date: 05/12/2024

Subject Name: Artificial Intelligence

Time: 2.30 PM To 5.30 PM

Total Marks: 100

Instructions:

1. Write down each question on separate page.
2. Attempt all questions.
3. Make suitable assumptions wherever necessary.
4. Figures to the right indicate full marks.

Q.1

Attempt any three.

Marks

- (a) Write a note on different types of Python Libraries : I. NumPy II Pandas III Matplot. Also give at least TWO functions/methods from each of the library. **08**
- (b) Bring out the difference between following: **08**
- 1) Classification vs Prediction
 - 2) Supervised Learning vs Unsupervised Learning
- (c) Consider the following set of points: $\{(30, 60), (40, 80), (50, 100)\}$. Find the least square regression line for the given data points. Using resultant regression line predict the data points. **08**

OR

Discuss in detail the various types of Machine learning models.

- (d) Explain in brief the different types of layers in an ANN? What is difference between single layer neural network and multi layer neural network **08**

Q.2

Attempt any three.

- (a) Mention and explain different Performance Metrics of the Supervised Learning ? **08**
- (b) List out and discuss different static as well as dynamic features, which can be used for malware analysis using Machine Learning **08**
- (c) List out and briefly explain types of Clustering. Find the SMC and Jaccard Coefficient for the following binary vector **08**
- X= 0101010001
Y=0100011000
- (d) Write a note on following: **08**
- 1) Overfitting (II) Under fitting (III) Image Segmentation (IV) Labelled vs Unlabeled Data

- Attempt any three.**
- Q.3 (a) Explain the working of a Convolutional Neural Network (CNN), describing its components and how it is used for image classification. 08
- (b) Illustrate the process with an example. 08
- (c) What is significance of data preprocessing phase in ML flow? What are the key steps involved in data preprocessing? 08
- (d) What are the limitations of Recurrent Neural Network (RNN)? How LSTM can overcome the limitations of RNN. Explain with the help of architecture of LSTM 08
- (d) How Decision Tree Algorithm works? List out the stopping criteria of Tree construction. Define Measure of Impurity. 07

- Attempt All.**
- Q.4 (a) Consider the following training data for the Naive Bayes Classifier.

chills	runny nose	headache	fever	flu?
Y	N	Mild	Y	N
Y	Y	No	N	Y
Y	N	Strong	Y	Y
N	Y	Mild	N	N
N	N	No	Y	Y
N	Y	Strong	N	N
N	Y	Strong	N	N
Y	Y	Mild	Y	Y

Which class is the candidate is classified in (Flu yes/ Flu No)? Show step by step execution.

Chills	Runny nose	Headache	Fever	flu
Y	N	Mild	Y	?

- (b) Explain the key steps involved in the Natural Language Processing (NLP) pipeline, providing examples of their application. 07

Q.5

Attempt All.

- (a) Define object tracking. List out and discuss types of object tracking. Explain object tracking algorithm with an example. 07
- (b) What is activation function? Explain different types of activation function in detail? 07

END OF PAPER

Seat No.: _____

Enrolment No. _____

NATIONAL FORENSIC SCIENCES UNIVERSITY

Semester End Examination (December – 2024)

M.Sc. Cyber Security Semester – III

Subject Code: CTMSCS SIII P5 E2

Subject Name: Network Forensics

Time: 10:30 AM-1:30 PM

Date: 10/12/2024

Total Marks: 100

Instructions:

1. Write down each question on a separate page.
2. Attempt all questions.
3. Make suitable assumptions wherever necessary.
4. Figures to the right indicate full marks.

			Marks
Q.1		Attempt any three.	
	(a)	Discuss different passive attacks devices for different transmission line.	08
	(b)	What are the main challenges in collecting network-based digital evidence? Discuss with relevant examples.	08
	(c)	Describe the main functionalities of a DNS and DHCP server and its relevance in network forensics investigations.	08
	(d)	How does Security Information and Event Management SIEM work? What are the challenges	08
Q.2		Attempt any three.	
	(a)	Differentiate between active and passive monitoring of network evidences.	08
	(b)	What is a firewall, and what role does it play in network security? Also explain its various types.	08
	(c)	Define different types of evidence and describe its importance in network forensics.	08
	(d)	How does packet analysis help in network forensics? Provide an overview of tools commonly used for this purpose.	08
Q.3		Attempt any three.	
	(a)	Discuss various attacks on WLAN and Countermeasures Against WLAN Attacks.	08
	(b)	Write a detailed note on NIDS/NIPS (Network Intrusion Detection/Prevention Systems). Include their functionality, types, and evidence they can provide in forensic investigations	08
	(c)	Describe the 802.11 protocol suite and its relevance in wireless forensics. Highlight the common attacks on wireless networks.	08
	(d)	During a routine network audit, an IT team discovers an unauthorized device generating significant traffic. The MAC address of the device does not match any known devices in the organization's inventory.	08

		Questions: 1. Explain how Address Resolution Protocol (ARP) and CAM table analysis can be used to identify and investigate the unauthorized device. 2. What actions should be taken to isolate the device and prevent further network compromise?	
Q.4		Attempt any two.	07
	(a)	Outline the process of wireless traffic capture and analysis. Discuss the challenges of investigating rogue wireless access points.	07
	(b)	Explain Penetration Testing Life Cycle.	
	(c)	An organization discovers an unauthorized wireless access point (WAP) within its premises, suspected to be a rogue access point used for an Evil Twin attack.	
		Questions: 1. Describe the process of identifying and locating the rogue WAP using wireless forensic techniques. 2. How would spectrum analysis and signal strength measurement assist in pinpointing the device? 3. What types of evidence can be collected from the rogue WAP, and how can this evidence be used in legal proceedings?	07
Q.5		Attempt any two.	
	(a)	What is the importance of logging in routers, switches, and firewalls during an investigation? Provide detailed examples of evidence that can be obtained from these devices.	07
	(b)	Outline the key principles of the OSCAR methodology in digital forensics and explain how they are implemented during an investigation	07
	(c)	A disgruntled employee is suspected of leaking confidential files to a competitor. Logs from the company's web proxy indicate unusual file uploads to external servers.	07
		Questions: 1. How would you analyze the web proxy logs to determine the employee's activity? 2. Discuss the types of evidence that could be gathered from web proxy servers and how this evidence could support the case. 3. Outline the steps to ensure chain of custody while collecting and analyzing evidence in this scenario.	

--- End of Paper---

NATIONAL FORENSIC SCIENCES UNIVERSITY

Semester End Examination (April - 2025)
M. Sc. Digital Forensics and Information Security
Semester - II

Subject Code: CTMSDFIS24 SH P1
Subject Name: Network Security & Forensic
Time: 10:30AM-1:30PM

Date: 24/04/2025

Total Marks: 100

Instructions:

1. Write down each question on a separate page.
2. Attempt all questions.
3. Make suitable assumptions wherever necessary.
4. Figures to the right indicate full marks.

		Marks
Q.1		
Attempt any three.		
(a)	Explain the OSI and TCP/IP models in detail. Compare their layers and functionalities.	08
(b)	Describe different perimeter devices like IDS, IPS, and Firewalls. How do they contribute to network security?	08
(c)	Explain the working of common network devices like switches, routers, and hubs with their functions in network topology.	08
(d)	Discuss different types of cyber-attacks such as ARP poisoning, MAC flooding, and DoS/DDoS.	08
Q.2		
Attempt any three.		
(a)	Explain the penetration testing life cycle. Describe each phase with real-world examples.	08
(b)	Discuss the types of scanning techniques used in penetration testing.	08
(c)	What is SNMP enumeration and how is it used in penetration testing?	08
(d)	Explain password cracking techniques and countermeasures.	08
Q.3		
Attempt any three.		
(a)	Describe symmetric and asymmetric encryption with examples.	08
(b)	Explain the concept of digital signatures and their verification process.	08
(c)	Write notes on hash functions and message digests. How are they used in security?	08
(d)	What is RSA algorithm. Given two prime numbers, $p=61$ and $q=53$, calculate the RSA public and private keys. ($e = 2$)	08
Q.4		
Attempt any two.		
(a)	Describe the 802.11 protocol and its security features.	07
(b)	What are WLAN attacks and how can they be prevented?	07
(c)	Explain DHCP and DNS Servers in the details.	07

Q.5	Attempt any two.	
(a)	What is digital evidence? Explain its types and importance in network forensics.	07
(b)	Describe the process of capturing and analyzing network traffic for forensic investigation.	07
(c)	Explain the key principles of the OSCAR methodology in digital forensic and its implementation during investigation.	07

--- End of Paper---

NATIONAL FORENSIC SCIENCES UNIVERSITY

M.Sc.

Semester End Examination (April - 2025)

Digital

Forensics

and

Information

Security

Semester - II

Subject Code: CTMSDFIS SII - P2**Subject Name:** Web Application Security**Time:** 10.30 AM - 01.30 PM**Date:** 25/04/2025**Total Marks:** 100**Instructions:**

1. Write down each question on a separate page.
2. Attempt all questions.
3. Make suitable assumptions wherever necessary.
4. Figures to the right indicate full marks.

- | | | Marks |
|------------|--|-------|
| Q.1 | Attempt any three. | |
| (a) | Explain NMAP and Its Scanning Techniques. | 08 |
| (b) | What is Threat Modeling? Explain any two Models. | 08 |
| (c) | What is Cross Site Scripting (XSS)? Explain how it can affect a website. | 08 |
| (d) | What do you understand by Sensitive Data Exposure? Give two examples of exposed data? | 08 |
| Q.2 | Attempt any three. | |
| (a) | Explain the lifecycle of a Vulnerability Assessment. Illustrate each phase with suitable examples? | 08 |
| (b) | What is a Web Service? Explain the difference between Web Service and Website? | 08 |
| (c) | What is IDOR ? Give a simple example. | 08 |
| (d) | Write a note on HTTP and its methods | 08 |
| Q.3 | Attempt any three. | |
| (a) | What are CVE and CWE? How are they useful in the context of vulnerability management? Provide relevant examples. | 08 |
| (b) | What is CMS? and Why CMS Security is important. | 08 |
| (c) | Explain File Upload Vulnerability. | 08 |
| (d) | What is Insecure Captcha? How can attackers bypass a weak CAPTCHA system? | 08 |
| Q.4 | Attempt any two. | |
| (a) | How do you handle a data breach during an incident? | 07 |
| (b) | Explain the role of Cookies and Sessions in Web Applications? | 07 |
| (c) | What is SQL Injection? Write a simple example to show how it works? | 07 |

Attempt any two.

Q.5

- (a) What is Docker? Write a note on Advantages of docker
- (b) What is Security Misconfiguration? Mention two common examples?
- (c) What are weak authentication tokens? Why are they risky in web applications?

--- End of Paper---